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Assignment No	:8
Roll No	:381
Batch	:E4
PRN No	:202502040032
Title of the Assignment	Java Web Frameworks:Spring

Problem Statement 1 : Write a program to implement MVC using Spring Framework.

Code:-

```
package com.mycompany.ead;

import org.springframework.context.annotation.*;
import org.springframework.jdbc.core.JdbcTemplate;
import org.springframework.jdbc.datasource.DriverManagerDataSource;
import org.springframework.stereotype.*;
import org.springframework.transaction.PlatformTransactionManager;
import org.springframework.transaction.annotation.*;
import org.springframework.web.bind.annotation.*;
import org.aspectj.lang.annotation.*;

import javax.sql.DataSource;

@Configuration
@ComponentScan(basePackages = "com.mycompany.ead")
@EnableAspectJAutoProxy
@EnableTransactionManagement
class AppConfig {

    @Bean
    public DataSource dataSource() {
        DriverManagerDataSource ds = new DriverManagerDataSource();
        ds.setDriverClassName("oracle.jdbc.driver.OracleDriver");
        ds.setUrl("jdbc:oracle:thin:@localhost:1521:xe");
        ds.setUsername("system");
        ds.setPassword("root");
        return ds;
    }
```

```java
    @Bean
    public JdbcTemplate jdbcTemplate(DataSource ds) {
        return new JdbcTemplate(ds);
    }

    // Transaction Manager (Important)
    @Bean
    public PlatformTransactionManager transactionManager(DataSource ds) {
        return new org.springframework.jdbc.datasource.DataSourceTransactionManager(ds);
    }
}

@Aspect
@Component
class LoggingAspect {

    // Runs before any method in EmployeeService
    @Before("execution(* com.mycompany.ead.EmployeeService.*(..))")
    public void logBefore() {
        System.out.println("\n[AOP NOTIFICATION]: Logging service activity...");
    }
}

@Service
class EmployeeService {

    private final JdbcTemplate jdbcTemplate;

    public EmployeeService(JdbcTemplate jdbcTemplate) {
        this.jdbcTemplate = jdbcTemplate;
    }

    @Transactional
    public void testDatabase() {

        // Simple query to test DB connection
        int count = jdbcTemplate.queryForObject(
                "SELECT count(*) FROM dual",
                Integer.class
        );
```

```java
        System.out.println(
                "Oracle Database Connection Successful! Count from dual: " + count
        );
    }
}

@RestController
@RequestMapping("/api")
class MyRestController {

    @GetMapping("/greet")
    public String sayHello() {
        return "Hello Yogesh! Your Spring REST Service is working.";
    }
}

public class SpringApp {

    public static void main(String[] args) {

        // Start Spring Container
        AnnotationConfigApplicationContext context =
                new AnnotationConfigApplicationContext(AppConfig.class);

        // Test AOP + JDBC
        EmployeeService service = context.getBean(EmployeeService.class);
        service.testDatabase();

        System.out.println("\nSpring Framework Practical initialized successfully!");

        context.close();
    }
}
```

Output:-